M24 Magnesium Sulphate

Scope of Practice	AP
Classification	• Electrolyte
Usual Presentation	• 5g in 10ml ampule • 500mg in 2ml vial
Mechanism of Action	• Reduces bronchial smooth muscle contraction resulting in bronchodilation. • Blocks calcium channels reducing calcium influx into cardiac myocytes, slowing abnormal electrical activity which causes Torsades de Pointes. • Dilates cerebral blood vessels, improving blood flow to the brain and reducing hypertension-related seizures, and as a central nervous system depressant, reducing neuronal excitability.
Indications	• Asthma (severe or life-threatening) • Eclampsia • Torsades de pointes
Contraindications	• Previous severe reaction to magnesium • Toxemia of pregnancy within the first 2 hours preceding delivery
Precautions	• Hypotension
Side Effects	• Arrhythmias • Hypotension (usually associated with fast administration) • Muscle weakness • CNS depression
Adult Dose & Administration	**Asthma** • 2g IV/IO/IM • IV/IO doses should be administered as an infusion over 20 minutes **Eclampsia** • 4g IV/IO over 5-10 minutes (dilute in 100ml NaCl or use syringe driver) • 8g IM (4mg into each gluteal muscle) **Torsades de Pointes** • 1-2g IV/IO over 5-20minutes • May repeat once if required
Pediatric Dose & Administration	**Asthma** • 40mg/kg IV/IO/IM **Torsades de Pointes** • 25-50mg/kg IV/IO over 5-20 minutes • May repeat once if required
Use in Pregnancy and Lactation	• Pregnancy: Safety has not been established. Magnesium should be administered if clinically indicated. • Lactation: Safety has not been established. Magnesium should be administered if clinically indicated.
Additional Information	• Do not run magnesium through the same line as adrenaline as precipitation may occur.